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CausalMixGPD: Bayesian Nonparametric Model with Optional Heavy-Tail Augmentaion and Causal Extensions

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Abstract

Heavy-tailed outcomes are commonly encountered in various scientific settings where rare extreme events dominate risk, cost, and policy recommendations. Although parametric models tend to adequately capture the center of a distribution, they often fail to characterize extreme outcomes, and non-parametric models lack principled extrapolation in tail regions. We develop `CausalMixGPD`, an R package for Bayesian modeling of heavy-tailed outcomes by integrating Dirichlet process mixtures for the center and generalized Pareto distributions for the tails in a unified framework. `CausalMixGPD` allows for both unconditional and covariate-dependent models, enabling flexible models of mixture weights and components as a function of predictors. Posterior computations were carried out using Markov chain Monte Carlo (MCMC) algorithms using the Nimble package in R. The framework is extended by allowing for separate modeling of outcome distributions for treatment groups. `CausalMixGPD` is designed for estimating and predicting causal quantities, such as average and quantile treatment effects, especially when quantiles are extreme. Moreover, this package provides a clustering framework for different kernel specifications. In R data analysis, it has demonstrated commendable performance in estimating extreme causal effects and clustering observations, thus providing a practical and extensible software framework for heavy-tailed outcome modeling.

Keywords: Bayesian nonparametrics; Dirichlet Process Mixture; Generalized Pareto Distribution; Extreme Value Theory; Cluster Analysis; Causal Inference; R; `nimble`.

1. Introduction

Causal inference is often summarized through average effects, but many applications require distributional comparisons between treatment regimes. In clinical trials, investigators may want to understand how a treatment changes outcomes across the outcome scale, including

effects among individuals in the upper tail where events are rare but consequences can be substantial. This shift from mean-based to distribution-based causal questions makes causal analysis tightly linked to how well we can estimate (and predict) outcome distributions under competing treatment assignments.

A growing R ecosystem supports Bayesian causal inference through flexible outcome modeling and posterior uncertainty quantification. For example, bcf implements Bayesian Causal Forests for binary treatments and continuous outcomes, providing posterior inference for both average and heterogeneous treatment effects (**?**). The package bartCause builds causal estimators around Bayesian additive regression trees, using BART as a regression engine for treatment-response modeling and causal effect estimation (**??**). Complementing tree-based approaches, BCEE implements a Bayesian model-averaging strategy for causal effect estimation with support for binary or continuous exposures and outcomes (**??**). A recurring practical difficulty in quantile-based causal reporting is the quantile crossing issue, where quantile curves at different probability levels can cross when they are estimated separately. To avoid this, CausalMixGPD directly models outcome densities and derives quantiles and other functionals from the posterior predictive distributions.

In the context of density estimation and conditional density estimation, Dirichlet Process Mixtures (DPM) provide a versatile nonparametric solution because they can capture multimodality and latent heterogeneity without fixing the number of mixing components in advance. In the experimental R ecosystem, Bayesian nonparametric mixture modeling is well supported by software based on Dirichlet process (DP) ideas. For example, dirichletprocess provides a flexible interface for specifying DPMs and fitting them by posterior simulation, yielding posterior inference for mixture weights, component labels, and predictive densities under user-chosen kernels (**?**). The package DPpackage offers a broad suite of Bayesian semi- and nonparametric models in R, including DP-based formulations for marginal and conditional densities, regression functions, and model-based predictive distributions (**?**). More recently, BNPmix emphasizes practical Bayesian nonparametric mixture approaches, providing posterior inference for clustering structure and density/regression estimation under DP and Pitman–Yor-type mixture specifications (**?**).

That clustering perspective is also important for the present package. Because DP mixtures induce random partitions through latent allocation variables, they naturally support model-based subgroup discovery in addition to density estimation. Bayesian cluster analysis is typically reported through label-invariant summaries such as pairwise co-clustering probabilities (Binder 1978) and representative partitions selected by Dahl's least-squares criterion (Dahl 2006). In regression settings, clustering itself can depend on predictors: dependent Dirichlet process constructions allow covariates to affect mixture weights, component-specific parameters, or both (MacEachern 1999, 2000; De Iorio, Müller, Rosner, and MacEachern 2004; Quintana, Müller, Jara, and MacEachern 2022), and predictor-dependent stick-breaking schemes provide a practical route for covariate-modulated cluster prevalence (Ren, Du, Carin, and Dunson 2011). Existing software supports parts of this workflow, but it typically focuses on bulk density estimation and clustering under a single kernel family without an explicit adjustment for heavy tailed data. As a result, applications that require both latent subgroup discovery and tail extrapolation still need additional modeling outside the standard Bayesian nonparametric mixture workflow.

However, it is important to note that estimating values far in the tail requires special care when data becomes scarce. Additionally, using a single model in the presence of heavy tail data

can distort the fit in the central region. In such cases extreme value theory (EVT) provides a principled framework for modeling and extrapolation of tail behavior (**?**). EVT characterizes the asymptotic distribution of threshold exceedances, which can be used to estimate tail probabilities and quantiles beyond the range of observed data (**?**). Bayesian implementations of EVT are available in R when posterior uncertainty quantification for tail quantities are required. For example, evdbayes provides Bayesian inference for standard extreme-value models via posterior simulation, enabling posterior and posterior-predictive summaries for extreme-event quantities (**?**). For peaks-over-threshold (POT) approaches, the package POT implements a focused collection of tools for GPD modeling and threshold-based tail analysis (**?**). Finally, for extreme quantile estimation and rare-event tail summaries, extremefit provides a published software implementation targeted specifically at extreme-quantile estimation (**?**).

This article introduces an R package (CausalMixGPD) that connects Bayesian nonparametric modeling with EVT, clustering, and robust causal analyses in the presence of covariates (confounders). The package provides three simultaneous capabilities:

- **Bayesian nonparametric (BNP) density modeling and prediction with optional GPD tail:** CausalMixGPD fits DPMs using a user-chosen kernel below a certain threshold (bulk part), and can optionally splice a generalized Pareto tail above the threshold (tail part) to enable the estimation of tail quantities and posterior predictive inference for rare-event summaries.

- **Clustering under predictor dependent mixtures:** A dedicated clustering interface for both bulk-only and spliced models, supporting weight dependence, parameter dependence, or both, and returning label-invariant clustering summaries through posterior similarity matrices and Dahl representative partitions (Binder 1978; Dahl 2006).

- **Causal estimation and prediction for randomized controlled trials (RCTs) and observational studies:** The final goal of this package is to provide causal estimands by modeling treatment-specific potential outcome densities and reporting contrasts as functionals of the posterior over densities. Framework supports both randomized and observational settings, with the latter using a flexible Bayesian propensity score model to adjust for confounding.

Overall, CausalMixGPD offers a unified workflow for density estimation, prediction, and causal reporting when tail behavior is of significance. The package is implemented in R with computationally intensive components coded in C++ and integrated via the `nimble` framework for model specification and posterior simulation (R Package Authors 2025b). The remainder of this article is organized as follows: Section 2 provides background on the key modeling components, including DPMs, GPDs,and the spliced bulk-tail construction. Section 3 gives an overview of the package's main functions and user interface. Section 4 and Section 5 illustrates the package's capabilities through applications to causal inference and clustering on R datasets, respectively. Finally, Section 6 concludes with a discussion of limitations and future directions.

2. Background and Preliminaries

Let the outcome–covariate pair be defined by

$$(Y, X) \in \mathbb{R} \times \mathbb{R}^p. \tag{1}$$

For a fixed covariate value x, the conditional cumulative distribution function (CDF) and density are

$$F(y \mid x) = \Pr\,(Y \leq y \mid X = x), \qquad f(y \mid x) = \frac{\partial}{\partial y} F(y \mid x), \tag{2}$$

and the conditional quantile function is defined by

$$Q(\tau \mid x) = \inf\Big\{y \in \mathbb{R} :\, F(y \mid x) \geq \tau\Big\}, \qquad 0 < \tau < 1. \tag{3}$$

CausalMixGPD treats conditional distribution of Y given X as the primary inferential object. The conditional density is modeled using a spliced (two part) construction that combines a Non–Parametric model for the center of the distribution where data is substantial and an extreme value distribution for the tail where data is scarce. All posterior summaries reported by the package are functionals of estimated conditional distributions' parameters, including conditional quantiles, densities, or samples.

The following subsections provide background on these two components and their integration into a spliced model.

2.1. Spliced DPM–GPD Conditional Distribution

CausalMixGPD adopts a spliced (bulk–tail) construction that retains the DPM fit below a high threshold and replaces the upper tail by a GPD component (Frigessi, Haug, and Rue 2002; Behrens, Lopes, and Gamerman 2004; Coles 2001). Let

$$p_u(x) = F_{\mathrm{DP}}(u(x) \mid x; \Theta) \tag{4}$$

denote the bulk probability mass below a threshold ($u(x)$) under the DPM fit. The spliced conditional CDF is given as

$$F(y \mid x; \Theta, \Phi) = \begin{cases} F_{\mathrm{DP}}(y \mid x; \Theta), & y \leq u(x), \\ p_u(x; \Theta) + \{1 - p_u(x; \Theta)\}\, F_{\mathrm{GPD}}(y \mid x; \Phi), & y > u(x), \end{cases} \tag{5}$$

where $F_{\mathrm{DP}}(\cdot)$ is the DPM CDF on the bulk region and $F_{\mathrm{GPD}}(\cdot)$ is the GPD CDF on the exceedance region (Section 7.1). It can be easily verified that the CDF (5) is continuous at $u(x)$ and differentiating (5) yields the spliced density fucntion given as

$$f(y \mid x; \Theta, \Phi) = \begin{cases} f_{\mathrm{DP}}(y \mid x; \Theta), & y \leq u(x), \\ \{1 - p_u(x; \Theta)\}\, f_{\mathrm{GPD}}(y \mid x; \Phi), & y > u(x), \end{cases} \tag{6}$$

where the tail density is a scaled GPD density, with scaling ensuring normalization of $f(\cdot \mid x; \Theta, \Phi)$. The following figure illustrates the spliced density construction, showing the bulk DPM density, the GPD tail density, and the resulting spliced density.

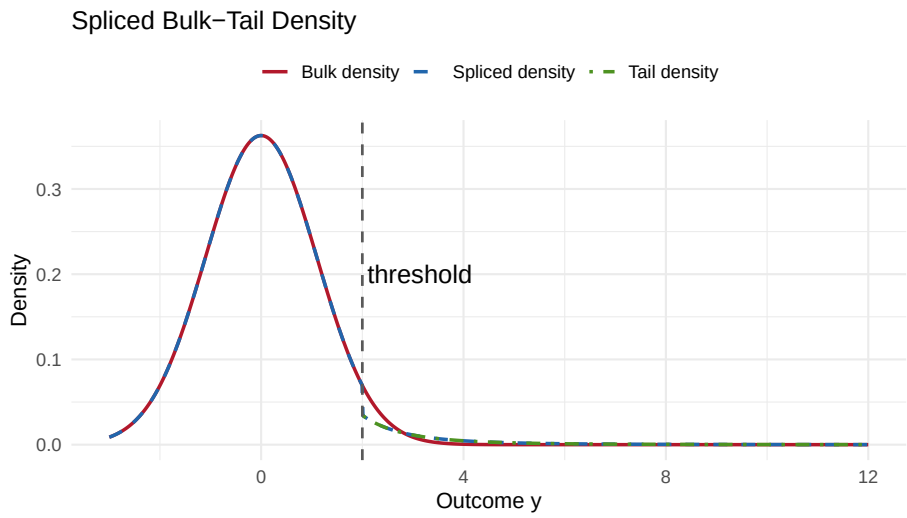

Figure 1: Illustration of the spliced bulk–tail density construction.

To find quantiles below the threshold i.e. for quantile index $\tau \in (0, p_u(x;\Theta)]$, one need to invert the CDF in (5) and obtain quantiles numerically. For $\tau \in (p_u(x;\Theta), 1)$, the index needs to be scaled in terms of threshold excedance probabilty which is defined as

$$\tilde{\tau}(x;\Theta) = \frac{\tau - p_u(x;\Theta)}{1 - p_u(x;\Theta)} \in (0,1), \tag{7}$$

and then the tail quantiles can be obtained by plugging in $\tilde{\tau}(x;\Theta)$ into the GPD quantile function in (20). Thus, the spliced quantile function is given by

$$Q(\tau \mid x;\Theta,\Phi) = \begin{cases} Q_{\mathrm{DP}}(\tau \mid x;\Theta), & \tau \le p_u(x;\Theta), \\ Q_{\mathrm{GPD}}(\tilde{\tau}(x;\Theta) \mid x;\Phi), & \tau > p_u(x;\Theta). \end{cases} \tag{8}$$

To make the spliced form and its associated functionals explicit, the parameters of the bulk and tail components are partitioned into two collections, denoted by Θ and Φ, respectively. Specifically, the bulk distribution defined in Equation (22) can be written as

$$f_{\mathrm{DP}}(y \mid x;\Theta) = \sum_{j=1}^{\infty} w_j \, k(y \mid x;\theta_j(x)) \,, \tag{9}$$

with $\Theta = \{\beta_j\}_{j\ge1}$, where β_j denotes a set of component-specific regression coefficients. The corresponding kernel parameter at covariate value x is given by

$$\theta_j(x) = g(x^\top \beta_j),$$

where $g(\cdot)$ is a user-specified link function. The collection Θ may also include kernel parameters that do not depend on covariates. For example, under a Gaussian kernel, Θ may include regression coefficients β_j^μ along with a component-specific variance parameter σ_j that does not depend on x.

For the tail component, define

$$\Phi = \{\beta_u, \beta_\sigma, \xi\}, \tag{10}$$

where the threshold and scale functions are specified as

$$u(x) = g_u(x^\top \beta_u), \qquad \sigma(x) = g_\sigma(x^\top \beta_\sigma),$$

with $g_u(\cdot)$ and $g_\sigma(\cdot)$ denoting user-specified link functions. The parameter ξ is the GPD shape parameter governing tail heaviness (see Section 7.1).

This package allows users to specify which parameters vary with covariates and which are treated as fixed or entirely determined by prior specifications (see Section 3).

2.2. Clustering under predictor-dependent Bayesian nonparametric mixtures

Predictor-dependent clustering is handled through Dependent Dirichlet Process (DDP) mixtures (MacEachern 1999, 2000), which index the random mixing distribution by covariates:

$$f(y \mid x) \;=\; \int k(y \mid x, \theta)\; dG_x(\theta) \;=\; \sum_{j=1}^{\infty} w_j(x)\, k(y \mid x, \theta_j)\,, \tag{11}$$

Here G_x is a predictor-indexed random measure, and its discreteness induces latent clusters. In CausalMixGPD, predictor dependence may enter through the weights, the component-specific parameters, or both (De Iorio *et al.* 2004; Ren *et al.* 2011; Quintana *et al.* 2022); clustering is reported through label-invariant summaries rather than raw MCMC labels (Binder 1978; Dahl 2006). Appendix Section 7.3 records the three supported dependence structures together with the PSM and representative-partition summaries.

2.3. Causal Inference: Notation, Identification, and Reported Estimands

We use the potential outcomes framework (Neyman 1990; Rubin 1974; Imbens and Rubin 2015). Let A denote treatment, X pre-treatment covariates, and $Y(a)$ the potential outcome under arm a. The observed outcome satisfies

$$Y = A\,Y(1) + (1 - A)\,Y(0). \tag{12}$$

The basic causal objects are the treatment-specific conditional distributions $F_a(y \mid x) = \Pr\{Y(a) \le y \mid X = x\}$ together with their densities, quantiles, and means. In randomized studies these are identified by design; in observational studies we additionally assume SUTVA/consistency, conditional ignorability $(Y(0), Y(1)) \perp A \mid X$, and overlap $0 < \Pr(A = 1 \mid X = x) < 1$ (Rosenbaum and Rubin 1983; Imbens and Rubin 2015). When needed, each arm-specific outcome model can be augmented with an estimated propensity score $\widehat{\rho}(x)$. Appendix Section 7.4 records the formal standardized functionals, identification assumptions, and propensity-score augmentation notation used in the causal extension.

The package reports the causal estimands in Table 1. Here superscripts m and t denote full-population and treated-population standardization, respectively.

Table 1: Causal estimands reported by the package

Estimand	Formula	Target population / conditioning
ATE	$\mu_1^{\mathrm{m}} - \mu_0^{\mathrm{m}}$	Population-standardized mean contrast
ATT	$\mu_1^{\mathrm{t}} - \mu_0^{\mathrm{t}}$	Treated-population mean contrast
QTE(τ)	$Q_1^{\mathrm{m}}(\tau) - Q_0^{\mathrm{m}}(\tau)$	Population-standardized τ-quantile contrast
QTT(τ)	$Q_1^{\mathrm{t}}(\tau) - Q_0^{\mathrm{t}}(\tau)$	Treated-population τ-quantile contrast
CATE(x)	$\mu_1(x) - \mu_0(x)$	Conditional mean contrast at covariate profile x
CQTE($\tau \mid x$)	$Q_1(\tau \mid x) - Q_0(\tau \mid x)$	Conditional τ-quantile contrast at covariate profile x

3. Package Overview: Workflow, Model, and Inference

This section describes the main workflow provided by CausalMixGPD for one-arm conditional modeling and prediction when a generalized Pareto tail is spliced above a covariate-dependent threshold. The causal extension reuses the same modeling, estimation, and prediction machinery after introducing treatment-specific outcome models and a propensity score component; we defer those additions to the causal section.

3.1. Augmented DPM–GPD model, likelihood, and priors

Let the observed data be

$$\mathcal{D} = \{(y_i, x_i)\}_{i=1}^n,$$

```
data("nc_posX100_p3_k2", package = "CausalMixGPD")
dat <- data.frame(y = nc_posX100_p3_k2$y, nc_posX100_p3_k2$X)
```

with $x_i \in \mathbb{R}^p$ denoting the covariate vector (including an intercept if desired). The target conditional distribution $F(\cdot \mid x)$ and its spliced DPM–GPD construction are defined in Section 2, with the spliced CDF and density given in (5)–(6). Posterior computation in CausalMixGPD approximates the bulk DPM by a finite truncation with J mixture components,

$$f_{\mathrm{DP}}(y \mid x; \Theta) \approx \sum_{j=1}^{J} w_j \, k(y \mid x; \theta_j), \qquad j = 1, \ldots, J,$$

where j indexes mixture components, $w_j \geq 0$, and $\sum_{j=1}^J w_j = 1$. Kernel parameters can be constant or covariate-dependent through link specifications of the form

$$g(\psi_j(x)) = x^\top \beta_{j,\psi}, \tag{13}$$

where $g(\cdot)$ enforces parameter support (e.g., positive scale parameters).

```
mcmc_fixed
```

```
fit <- dpmgpd(
  formula = y ~ x1 + x2 + x3,
  data = dat,
  backend = "sb",
  kernel = "lognormal",
  components = 5,
  mcmc = mcmc_fixed
)
```

For the spliced model, let $u(x)$ be the threshold, $\sigma(x) > 0$ the tail scale, and $\xi \in \mathbb{R}$ the tail shape (as in Section 2.1). Define $\delta_i = \mathbf{1}\{y_i > u(x_i)\}$. Using the spliced density (6) with the truncated bulk mixture, the likelihood is

$$L(\Theta, \Phi; \mathcal{D}) = \prod_{i=1}^{n} f(y_i \mid x_i; \Theta, \Phi), \tag{14}$$

where Θ collects the truncation-level bulk parameters $\{w_j, \theta_j\}_{j=1}^{J}$, and Φ collects the tail parameters governing $u(x)$, $\sigma(x)$, and ξ. Equivalently, the log-likelihood can be written as

$$\begin{aligned} \log L(\Theta, \Phi; \mathcal{D}) \;=\; & \sum_{i=1}^{n} (1 - \delta_i) \log f_{\mathrm{DP}}(y_i \mid x_i; \Theta) \\ & + \sum_{i=1}^{n} \delta_i \Big[\log\{1 - F_{\mathrm{DP}}(u(x_i) \mid x_i; \Theta)\} + \log f_{\mathrm{GPD}}(y_i \mid x_i; \Phi) \Big], \end{aligned} \tag{15}$$

where f_{GPD} is given in (19) and $F_{\mathrm{DP}}(\cdot \mid x)$ is the bulk mixture CDF induced by the truncated mixture.

Priors follow the decomposition implied by (23) and (24). Under the SB backend, the truncation-level weights are generated through Beta stick breaks (Section 2, (23)), which are represented in nimble using the Beta distribution in model code (see `dbeta`). Under the CRP backend, the latent-cluster-label representation is implemented using the CRP distribution in nimble (see `dCRP`), matching the predictive representation in (24). Kernel parameters and tail parameters are assigned weakly informative priors, with regression coefficients $\beta_{j,\psi}$ arising through the link representation (13). The same link construction is used for covariate-dependent $u(x)$ and $\sigma(x)$ when those are modeled as functions of x.

3.2. MCMC algorithm, convergence diagnostics, and parameter extraction

Posterior simulation targets $p(\Theta, \Phi \mid \mathcal{D}) \propto L(\Theta, \Phi; \mathcal{D})\, p(\Theta, \Phi)$ using the likelihood (14) and the prior structure implied by the SB or CRP representation (Section 2, (23)–(24)). Given a model specification, CausalMixGPD runs MCMC through nimble (de Valpine, Turek, Paciorek, Anderson-Bergman, Temple Lang, and Bodik 2017; R Package Authors 2025b) under the chosen backend; SB uses Beta stick breaks in model code (see `dbeta`) and CRP uses the CRP distribution for latent cluster labels (see `dCRP`). Users control the number of iterations, burn-in, thinning, and number of chains through the `mcmc` argument in the wrappers.

Convergence checking is supported through standard multi-chain diagnostics and trace-based visualizations. The model object supports `summary()` for posterior summaries, `plot()` for

diagnostic plots (trace, running mean, autocorrelation, and multi-chain diagnostics such as $\widehat{R}$ when multiple chains are used), and `params()` as a lightweight extractor returning posterior mean parameters reshaped to natural dimensions. Interval summaries used by `predict()` can be chosen as equal-tailed credible intervals or HPD intervals as described in Section 3.3 (Gelman, Carlin, Stern, Dunson, Vehtari, and Rubin 2013; Plummer, Best, Cowles, and Vines 2024). Diagnostic plotting relies on established MCMC diagnostic tooling (R Package Authors 2025a).

```
print(fit)
p_hat <- params(fit)
p_hat
```

Bulk-only special case (no GPD tail)

Setting `GPD=FALSE` removes the splicing step so that $f(y \mid x; \Theta) \equiv f_{\mathrm{DP}}(y \mid x; \Theta)$, and $L(\Theta, \Phi; \mathcal{D})$ reduces to the bulk-mixture likelihood in Θ. All estimation and prediction targets described below remain available, with tail-specific terms omitted. The wrapper `dpmix()` fits this bulk-only model, while `dpmgpd()` fits the augmented (spliced) model.

3.3. Posterior functionals, interval summaries and prediction

The estimands reported by CausalMixGPD are functionals of the estimated conditional distribution $F(\cdot \mid x)$ from Section 2. In this article we focus on four conditional functionals: density $f(y \mid x)$, survival $S(y \mid x) = 1 - F(y \mid x)$, quantiles $Q(\tau \mid x)$, and mean-type summaries.

Let $(\Theta^{(s)}, \Phi^{(s)})_{s=1}^{S}$ denote post-burn-in draws from the posterior. For density, survival, and quantiles, CausalMixGPD evaluates the corresponding draw-wise functional using the spliced formulas (5)–(8) (with the truncated bulk mixture), and then summarizes across draws. Specifically, for a generic scalar functional $T(\Theta, \Phi; x)$ (e.g., $f(y_0 \mid x)$ or $Q(\tau \mid x)$),

$$\widehat{T}(x) = \frac{1}{S} \sum_{s=1}^{S} T(\Theta^{(s)}, \Phi^{(s)}; x),$$

and uncertainty is summarized either by equal-tailed credible intervals (posterior quantiles) or by highest posterior density (HPD) intervals (shortest intervals covering the target posterior mass). Equal-tailed intervals and general Bayesian posterior summaries follow standard practice (Gelman *et al.* 2013); HPD intervals are computed from the MCMC draws using established implementations (Plummer *et al.* 2024).

Mean-type summaries are handled differently in the package. For `type="mean"` and `type="rmean"`, CausalMixGPD uses posterior predictive Monte Carlo within each posterior draw rather than direct plug-in evaluation. For a fixed covariate value x, within draw s the package generates

$$y_1^{(s)}, \ldots, y_M^{(s)} \;\sim\; f(\cdot \mid x; \Theta^{(s)}, \Phi^{(s)}),$$

and computes

$$\mu^{(s)}(x) = \frac{1}{M} \sum_{m=1}^{M} y_m^{(s)}, \qquad \mu_c^{(s)}(x) = \frac{1}{M} \sum_{m=1}^{M} \min\{y_m^{(s)}, c\},$$

corresponding to the conditional mean and the restricted mean at cutoff c, respectively. Posterior summaries and intervals are then formed from $\{\mu^{(s)}(x)\}_{s=1}^{S}$ or $\{\mu_c^{(s)}(x)\}_{s=1}^{S}$. Under the spliced model, the conditional mean is infinite when $\xi \geq 1$; accordingly, CausalMixGPD flags `type="mean"` as infinite when there is posterior mass with $\xi \geq 1$, while `type="rmean"` remains finite by construction.

Using the same fitted model `fit` from Section 3.1, we now compute posterior summaries on the original training covariates (default behavior of `predict()` when `newdata` is not supplied).

```
q_grid <- c(0.50, 0.90, 0.95, 0.99)
x_eval <- dat[1:20, c("x1", "x2", "x3")]

est_quant <- predict(fit, newdata = x_eval, type = "quantile", index = q_grid,
                     interval = "hpd", level = 0.95)
```

For a new covariate value $x^\star$, prediction is based on the posterior predictive distribution

$$p(y^\star \mid x^\star, \mathcal{D}) = \int f(y^\star \mid x^\star; \Theta, \Phi)\, p(\Theta, \Phi \mid \mathcal{D})\, d\Theta\, d\Phi.$$

The `predict()` method returns posterior summaries for the same functional targets as in Section 3.3. For `type="density"` and `type="survival"`, the user supplies a grid of y values and the method returns pointwise posterior means with interval bounds computed draw-by-draw. For `type="quantile"`, the user supplies quantile indices τ and the method returns posterior summaries of $Q(\tau \mid x^\star)$ computed using the spliced quantile representation in (8). For `type="mean"` and `type="rmean"`, prediction uses posterior predictive Monte Carlo as described above, with x replaced by $x^\star$ and with user-controlled M through `nsim_mean`.

```
q_levels <- c(0.25, 0.50, 0.75)
x_new <- as.data.frame(lapply(
  dat[, c("x1", "x2", "x3")],
  quantile,
  probs = q_levels,
  na.rm = TRUE
))
rownames(x_new) <- c("q25", "q50", "q75")
y_grid <- seq(0, 10, length.out = 200)
p_grid <- c(0.50, 0.90, 0.95, 0.99)
```

```
pdens <- predict(fit, newdata = x_new, y = y_grid, type = "density",
                 interval = "credible", level = 0.95)
```

```
psurv <- predict(fit, newdata = x_new, y = y_grid, type = "survival",
                 interval = "credible", level = 0.95)
```

```
pquant <- predict(fit, newdata = x_new, type = "quantile", index = p_grid,
                  interval = "hpd", level = 0.95)
```

3.4. Clustering extension

The clustering wrappers reinterpret the fitted mixture as a latent partition and return cluster summaries rather than only density functionals. `dpmix.cluster()` fits the bulk-only model and `dpmgpd.cluster()` adds the optional GPD tail, with predictor dependence entering through the weights, component parameters, or both.

Since mixture labels are permutation-invariant, the main outputs are label-invariant summaries: the posterior similarity matrix (PSM), a representative partition, cluster sizes, and assignment scores (Binder 1978; Dahl 2006). The same `predict()` method handles training labels and new-observation assignment. Appendix Section 7.3 gives the three dependence patterns and the PSM-based summary definitions.

```
data("nc_realX100_p3_k2", package = "CausalMixGPD")

dat_cl <- data.frame(y = nc_realX100_p3_k2$y, nc_realX100_p3_k2$X)

fit_cluster <- dpmix.cluster(
  y ~ x1 + x2 + x3,
  data       = dat_cl,
  kernel     = "normal",
  type       = "param",
  components = 8,
  mcmc = mcmc_fixed
)
```

```
cluster_psm   <- predict(fit_cluster, type = "psm")
cluster_train <- predict(fit_cluster, type = "label", return_scores = TRUE)
cluster_new   <- predict(
  fit_cluster,
  newdata = dat_cl,
  type = "label",
  return_scores = TRUE
)

cluster_sizes <- data.frame(
  cluster = names(summary(cluster_train)$cluster_sizes),
  n = as.integer(summary(cluster_train)$cluster_sizes)
)
cluster_certainty <- apply(cluster_train$scores, 1, max)
cluster_new_certainty <- apply(cluster_new$scores, 1, max)
```

For heavy-tailed responses, the same workflow uses `dpmgpd.cluster()` in place of `dpmix.cluster()`.

3.5. Causal extension and propensity score augmentation

The causal wrappers fit separate treated and control outcome models and can augment the outcome covariates with a propensity-score (PS) summary. With PS augmentation,

$$A_i \mid \boldsymbol{x}_i \sim \text{Bernoulli}(e_{\boldsymbol{\gamma}}(\boldsymbol{x}_i)), \qquad \boldsymbol{r}(\boldsymbol{x}) = \Big(1,\ \boldsymbol{x}^\top,\ \psi(\widehat{e}(\boldsymbol{x}))\Big)^\top,$$

where the PS link is logit or probit and `ps_scale` controls $\psi(\cdot)$. The treated and control outcome models then reuse the same mixture-regression and optional GPD-tail structure

as the one-arm workflow; Appendix Section 7.4 gives the formal standardized functionals, identification assumptions, and PS-augmentation notation.

Fitting uses `mcmc_ps` for the PS model and `mcmc_outcome` for the arm-specific outcomes. Posterior contrasts are then summarized through `ate()`, `att()`, `qte()`, `qtt()`, `cate()`, and `cqte()`, with restricted-mean variants available when heavy tails make ordinary means unstable. For `predict()`, mean and quantile requests return treated-minus-control summaries, while density, survival, and tail probabilities are returned by arm. A minimal workflow is:

```
data("causal_pos500_p3_k2", package = "CausalMixGPD")
dat <- causal_pos500_p3_k2
df  <- data.frame(y = dat$y, A = dat$A, dat$X)
```

```
cfit <- dpmgpd.causal(
  formula = y ~ x1 + x2 + x3,
  data = df,
  treat = "A",
  backend = "sb",
  kernel = "lognormal",
  components = 5,
  PS = "logit",
  ps_scale = "logit",
  ps_summary = "mean",
  mcmc_outcome = mcmc_fixed,
  mcmc_ps = mcmc_fixed
)
```

```
ate_hat <- ate(cfit, level = 0.90, interval = "credible")
qte_hat <- qte(cfit, probs = c(0.50, 0.90, 0.95), level = 0.90, interval = "hpd")
qs    <- c(0.25, 0.50, 0.75)
Xgrid <- expand.grid(lapply(df[c("x1","x2","x3")], quantile, probs = qs))
pred_q <- predict(cfit, newdata = Xgrid, type = "quantile", p = c(0.50, 0.90))
pred_s <- predict(cfit, newdata = Xgrid, type = "survival", y = rep(4, nrow(Xgrid)))
```

The bulk-only analogue is `dpmix.causal()`, which keeps the same causal estimand and prediction interface without the GPD tail.

4. Data analysis I: Clustering

This application demonstrates the clustering wing for a conditional mixture model under a bulk-only likelihood (no GPD tail). The objective is to infer latent groups of observational units implied by the fitted mixture model, and to summarize posterior clustering uncertainty in a label-invariant way. Throughout, clustering is reported via (i) the posterior similarity matrix (PSM) and (ii) a single representative partition derived from the posterior draws, together with out-of-sample cluster label prediction for new units.

4.1. Dataset and objective

We use the `Boston` housing data (MASS), treating each census tract as an observational unit. Let

$$\mathcal{D} = \{(y_i, \boldsymbol{x}_i)\}_{i=1}^n,$$

where y_i is the per-capita crime rate (`crim`) for tract i, and $\boldsymbol{x}_i$ is a vector of tract characteristics. Our aim is twofold: (i) to cluster tracts using a flexible conditional mixture model that can accommodate skewness and heterogeneity, and (ii) to assign cluster labels to new tracts using posterior predictive co-clustering evidence.

4.2. Preprocessing and train/test split

We split the data into a training set used to fit the clustering model and a test set used only for cluster-label prediction. Covariates are standardized using training-set moments to avoid leakage.

```
library(MASS)

set.seed(123)
data("Boston", package = "MASS")

dat <- Boston
dat <- subset(dat, select = c(crim, lstat, rm, nox))

n <- nrow(dat)
idx_train <- sample(seq_len(n), size = floor(0.75 * n), replace = FALSE)
train_dat <- dat[idx_train, ]
test_dat  <- dat[-idx_train, ]

# standardize covariates using training set
xvars <- c("lstat", "rm", "nox")
mu_x  <- vapply(train_dat[, xvars], mean, numeric(1))
sd_x  <- vapply(train_dat[, xvars], sd,   numeric(1))

train_dat[, xvars] <- sweep(sweep(train_dat[, xvars], 2, mu_x, "-"), 2, sd_x, "/")
test_dat[, xvars]  <- sweep(sweep(test_dat[, xvars],  2, mu_x, "-"), 2, sd_x, "/")
```

```
  sample   n
1  train 379
2   test 127
```

4.3. Clustering model specification and fit

We fit a bulk-only conditional mixture model for `crim` with three covariates. In this illustration, we select `type="param"`, meaning covariates enter through component-specific regression parameters while the mixture weights are global. This configuration clusters observations by their regression behavior and distributional differences while keeping cluster prevalence interpretable as a global mixing proportion.

The fitted object is subsequently summarized through a representative partition together with cluster-specific response summaries.

```
fit_clust <- dpmgpd.cluster(
  formula    = crim ~ lstat + rm + nox,
  data       = train_dat,
  kernel     = "lognormal",
  type       = "weights",
```

```
  components = 15,
  mcmc       = mcmc_fixed
)
```

4.4. Representative partition and cluster-specific response summaries

Because mixture components are not identifiable up to permutation, the training labels are still extracted from the posterior through the PSM and Dahl's least-squares representative partition. For presentation, however, we focus on the resulting partition and the response distributions within its representative clusters rather than on the full PSM heatmap.

The following command extracts the representative labels for the training sample.

```
z_train <- predict(fit_clust, type = "label")
```

For presentation we summarize the first five most populated clusters from `summary(fit_clust,top_n=5)` and report the associated descriptive statistics in Table 3.

4.5. Cluster labels for new observations

Cluster assignment for new observations is performed through the same prediction interface. The output is a label in the space of the representative training clusters, obtained by aggregating posterior predictive co-clustering evidence at the cluster level.

```
z_test <- predict(
  fit_clust,
  newdata = test_dat,
  type    = "label"
)
```

For the test sample we tabulate only the predicted cluster sizes that are actually represented in the new observations (Table 2).

Table 2: Test cluster sizes

Cluster	C3	C2	C4	C7	C1	C8	C5
n	46	45	23	8	2	2	1

To summarize the dominant representative clusters, Table 3 reports descriptive statistics for up to the five most populated training clusters. The table is displayed with summary metrics in rows and retained clusters in columns, so the cluster sizes, within-cluster means, and within-cluster standard deviations can be compared across clusters directly.

Table 3: Descriptive summaries for up to the five most populated representative clusters

Metric	C3	C4	C8	C2	C7
n	75.000	68.000	47.000	45.000	35.000
crim mean	0.338	0.530	3.843	9.732	7.642
crim sd	1.066	3.464	7.858	16.201	15.235
lstat mean	-0.873	-0.398	-0.136	0.770	0.303
lstat sd	0.373	0.525	0.654	0.823	0.852
rm mean	0.846	-0.159	-0.276	-0.346	0.388
rm sd	0.831	0.564	0.545	0.510	0.864
nox mean	-0.763	-0.815	0.364	0.795	0.563
nox sd	0.463	0.431	0.914	0.802	0.874
certainty mean	0.169	0.144	0.128	0.131	0.147
certainty sd	0.030	0.028	0.028	0.028	0.045

This analysis illustrates the intended clustering workflow: inference is summarized through a representative training partition and its cluster-level descriptive summaries, and new observations are assigned to those clusters using posterior predictive co-clustering evidence. Tables 2 and 3 summarize the predicted test-cluster allocation and the dominant representative training clusters without additional data-analysis plots. In Section 5, we next turn to the causal workflow, which targets treatment contrasts rather than latent partitions.

5. Data analysis II: Causal Inference

This application demonstrates causal estimands from a two-arm spliced DPM regression with a GPD tail. In contrast to Section 4, where the focus is model-based clustering under a CRP bulk-only regression, here the focus is inference on restricted-mean and quantile treatment contrasts, including conditional (covariate-specific) effects and extreme upper-tail behavior.

5.1. Dataset and objective

We use the real `Treatment` dataset from `Ecdat`, which records training assignment, baseline covariates, and post-program earnings for a job-training evaluation study (LaLonde 1986; Dehejia and Wahba 1999, 2002). To keep vignette runtime modest while preserving the treated group, we use all treated units together with a reproducible random sample of 400 controls.

Let Y denote post-program earnings (USD), $A \in \{0, 1\}$ denote assignment to training, and $\boldsymbol{x}$ denote baseline covariates including age, education, ethnicity, marital status, previous earnings, and unemployment indicators. The scientific goal is to quantify how training changes (i) upper-tail-truncated average earnings and (ii) distributional features of earnings, with special attention to the upper tail.

For this heavy-tailed benchmark, unrestricted mean effects are sensitive to a small number of extreme posterior draws. I therefore use restricted-mean versions of the mean-based causal

estimands with cutoff c equal to the empirical 95th percentile of observed earnings. Throughout, we report the four causal estimands in Table 4. Here $Q_a(\tau \mid \boldsymbol{x})$ denotes the arm-specific τ-quantile. The standardized estimands (RMATE, QTE) integrate over the empirical distribution of $\{\boldsymbol{x}_i\}_{i=1}^n$, while the conditional estimands (RMCATE, CQTE) are evaluated at user-chosen covariate profiles.

Table 4: Causal estimands reported in the earnings application

Estimand	Definition	Evaluation
RMATE	$\mathbb{E}[\min\{Y(1), c\}] - \mathbb{E}[\min\{Y(0), c\}]$	Standardized over the empirical covariate distribution
QTE	$Q_{Y(1)}(\tau) - Q_{Y(0)}(\tau)$	Standardized over the empirical covariate distribution
RMCATE	$\mathbb{E}[\min\{Y(1), c\} \mid \boldsymbol{x}] - \mathbb{E}[\min\{Y(0), c\} \mid \boldsymbol{x}]$	Evaluated at representative covariate profiles
CQTE	$Q_1(\tau \mid \boldsymbol{x}) - Q_0(\tau \mid \boldsymbol{x})$	Evaluated at representative covariate profiles

The following chunk loads the data and constructs the design matrix used in the causal fit. We display the first six rows in the PDF for reference.

```r
library(Ecdat)

data("Treatment", package = "Ecdat")
set.seed(123)
trt_app2 <- Treatment[Treatment$treat, ]
ctl_app2 <- Treatment[!Treatment$treat, ]
ctl_app2 <- ctl_app2[sample(seq_len(nrow(ctl_app2)), 400), ]
dat_app2 <- subset(
  rbind(trt_app2, ctl_app2),
  select = c(treat, age, educ, ethn, married, re74, re75, re78, u74, u75)
)

y <- dat_app2$re78 + 1
A <- as.integer(dat_app2$treat)
X <- stats::model.matrix(
  ~ age + educ + ethn + married + re74 + re75 + u74 + u75,
  data = dat_app2
)[, -1, drop = FALSE]
covars <- colnames(X)
rm_cutoff <- as.numeric(stats::quantile(y, 0.95, na.rm = TRUE))

utils::head(dat_app2)
```

```
  treat age educ  ethn married re74 re75     re78  u74  u75
1  TRUE  37   11 black    TRUE    0    0  9930.05 TRUE TRUE
2  TRUE  30   12 black   FALSE    0    0 24909.50 TRUE TRUE
3  TRUE  27   11 black   FALSE    0    0  7506.15 TRUE TRUE
4  TRUE  33    8 black   FALSE    0    0   289.79 TRUE TRUE
5  TRUE  22    9 black   FALSE    0    0  4056.49 TRUE TRUE
6  TRUE  23   12 black   FALSE    0    0     0.00 TRUE TRUE
```

5.2. Spliced DPM regression model

I model arm-specific conditional outcome distributions $F_a(\cdot \mid \boldsymbol{x})$ via a spliced construction:

$$f_a(y \mid \boldsymbol{x}) = \sum_j w_{aj}(\boldsymbol{x})\, k(y \mid \Theta_{aj})\, \mathbf{1}(y \le u) + \{1 - F_a(u \mid \boldsymbol{x})\}\, g(y \mid u, \sigma_a, \xi_a)\, \mathbf{1}(y > u),$$

where the bulk kernel $k(\cdot)$ is a DPM regression (stick-breaking backend) and the tail density $g(\cdot)$ is GPD with scale $\sigma_a > 0$ and shape ξ_a. The GPD tail index ξ_a controls extremal heaviness (e.g., $\xi_a > 0$ corresponds to heavy tails). This model stabilizes inference for extreme quantiles by coupling nonparametric bulk flexibility with EVT-consistent tail extrapolation.

5.3. Model fitting

I fit a two-arm spliced mixture using a lognormal bulk kernel after shifting earnings by one dollar, so that zero post-program earnings remain compatible with the positive-support bulk model while leaving treatment contrasts unchanged. The MCMC settings below are intentionally modest for vignette speed; in substantive analysis we recommend longer runs and sensitivity checks (e.g., components, threshold priors, and prior scales).

```
fit <- dpmgpd.causal(
  x = y,
  X = X,
  treat = A,
  backend = "sb",
  kernel = "lognormal",
  components = 4,
  mcmc = c(mcmc_fixed, list(show_progress = FALSE))
)
```

5.4. Standardized estimands: QTE

I summarize the overall effect using QTE on a quantile grid that extends into the upper tail but stops short of the most unstable 0.99 level in the main text.

```
qte_fit <- qte(fit, probs = taus, show_progress = FALSE)
```

Table 5: Marginal QTE estimates across the main-text quantile grid

index	estimate	lower	upper
0.25	0.176	0.01	0.963
0.50	-20.273	-53.56	-2.855
0.75	-24507.906	-39306.07	-19120.043
0.90	-212710.523	-2187278.40	832145.875
0.95	-15744936.770	-110430994.10	53723995.298

Table 5 reports the marginal quantile treatment-effect summaries for the spliced causal model. This table replaces the earlier effect plots in the data-analysis presentation.

5.5. Conditional estimands: restricted mean CATE and CQTE at representative profiles

To illustrate heterogeneity, we evaluate conditional effects at two covariate profiles: (i) a "typical" profile at sample medians and (ii) a "high previous earnings" profile at the 80th percentile of 1975 earnings, holding other covariates at medians.

```
X_profiles <- if (app2_cache_matches_data(fit, X)) {
  X
} else {
  fit_ps_x <- fit$bundle$design$data$X
  fit_ps_x[, setdiff(colnames(fit_ps_x), "(Intercept)"), drop = FALSE]
}

col_med <- apply(X_profiles, 2, median, na.rm = TRUE)
x_typical <- col_med
x_highprev <- col_med
profile_var <- if ("prevearn" %in% colnames(X_profiles)) {
  "prevearn"
} else if ("re75" %in% colnames(X_profiles)) {
  "re75"
} else if ("re74" %in% colnames(X_profiles)) {
  "re74"
} else {
  colnames(X_profiles)[1]
}
x_highprev[profile_var] <- as.numeric(
  quantile(X_profiles[, profile_var], 0.80, na.rm = TRUE)
)
xnew <- rbind(typical = x_typical, high_prev_earn = x_highprev)
profile_names <- rownames(xnew)
profile_names <- gsub("_", " ", profile_names, fixed = TRUE)
profile_labels <- sprintf("id = %d (%s)", seq_len(nrow(xnew)), profile_names)
```

Table 6: Representative profiles used for conditional causal contrasts; the high-profile shift is applied to re75.

	age	educ	ethn_blk	ethn_hisp	married	re74	re75	u74	u75
typical	29	12	0	0	1	13244.6	10741.9	0	0
high prev earn	29	12	0	0	1	13244.6	24706.5	0	0

```
cate_fit <- cate(
  fit,
  newdata = xnew,
  type = "rmean",
  cutoff = rm_cutoff,
  show_progress = FALSE
)
```

Table 7: Restricted-mean CATE estimates with cutoff c = 41377.300

id	estimate	lower	upper
1	-4727.590	-8532.220	-1246.541
2	-3672.043	-9618.523	1400.155

```
cqte_fit <- cqte(fit, probs = taus, newdata = xnew, show_progress = FALSE)
```

Table 8: Conditional QTE estimates at the representative profiles

	id = 1 (typical)			id = 2 (high prev earn)		
τ	Estimate	Lower	Upper	Estimate	Lower	Upper
0.25	0.033	0.005	0.200	0.033	0.005	0.200
0.50	0.000	0.000	0.000	0.000	0.000	0.000
0.75	-1329.662	-2108.182	-719.495	-1329.662	-2108.182	-719.495
0.90	-19894.435	-32604.688	-14853.863	47655.316	-80646.105	378512.884
0.95	-15470.430	-68346.394	1915.215	228588.651	-285055.926	1358973.126

Tables 7 and 8 summarize the conditional restricted-mean and quantile treatment effects at the representative covariate profiles.

5.6. Tail behavior

A key advantage of the spliced model is direct posterior inference on the tail index ξ_a by arm. Larger ξ_a indicates a heavier upper tail and greater propensity for extreme earnings.

Table 9: Posterior summaries of the arm-specific GPD tail-shape parameters

arm	mean	sd	q0.025	q0.500	q0.975
treated	0.292	0.117	0.055	0.289	0.477
control	0.520	0.345	0.212	0.460	1.325

5.7. Interpretation

The QTE profile shows that the treatment effect strengthens markedly in the upper part of the outcome distribution. The restricted-mean CATE estimates are positive at both representative profiles, while the CQTE curves show stronger upper-tail gains for the higher previous-earnings profile. Finally, the treated arm has the larger posterior tail index ξ_a, indicating a heavier upper tail under training, which is consistent with stronger upper-tail treatment gains in this application.

6. Discussion

CausalMixGPD combines the flexibility of central distribution modeling, the rigor of upper-tail extrapolation, and the ability to compare the full outcome distributions for causal inference in a unified framework. It allows the full flexibility of the non-parametric bulk, the addition of GPD structure in the tails, and the ability to perform a unified inference for the posterior summaries.

The article focuses on the semiparametric aspect of the model, but the software has a wider scope that allows the user to perform a variety of tasks beyond the GPD splicing. It allows the

user to perform stick-breaking and Chinese restaurant processes, different kernel families for real-line and positive outcomes, arm-specific causal inference, GPD splicing, and a clustering framework that uses the same machinery as the mixture fitting.

The clustering framework allows the dependence to be modeled through the weights, the atoms, or a combination of both, which is consistent with the dependent DP regression clustering framework. The posterior uncertainty in the clustering partitions is also performed by displaying the similarity matrices and the representative partitions, making the clustering task interpretable even in the presence of label switching in the MCMC samples.

The clustering, bulk tail, and posterior predictive analyses are all connected in a single workflow. This allows for a smooth transition from exploratory subgroup analyses to tail risk assessment and causal inference.

Some Limitations. One limitation of this method is that it is computationally expensive. MCMC for mixture models with covariate-dependent parameters and tail components is computationally expensive, especially when fitting separate chains for each arm, including separate models for each outcome, and including a series of PS estimation models. Larger sample sizes and more dimensions will be even more expensive, especially for sensitivity analyses and model comparison.

Another limitation of this method is that it is a spliced model, and this has all of the tuning parameters of a Dirichlet process mixture and a PoT model. There is a degree of uncertainty in the results, depending on the choice of kernel, truncation, and tail parameters. GPD is available for EVT for high thresholds, but for finite sample tail estimation, there must be sufficient exceedances. Similarly, clustering in a spliced model has similar behavior: strong overlap, weak predictor effects, and/or insufficient exceedances all contribute to a high degree of uncertainty in clustering, where the posterior similarity matrix is more informative than a clustering output.

The causal implementation is currently limited to binary treatments and continuous outcomes, assuming standard identification conditions of consistency, ignorability, and overlap. Propensity score augmentation is helpful for adjustments in observational studies, but it is no replacement for good study design, overlap, and sensitivity analyses for unmeasured confounding. It is not a framework for censored survival, multivariate outcomes, longitudinal treatments, or interference.

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7. Appendix: Additional Theoretical Background and Supplementary Details

This appendix keeps only the supporting theory cited from the main text, including the fuller clustering and causal formulations deferred from Section 2.

7.1. Generalized Pareto Distribution for Threshold Exceedances

In modeling terms, the tail region is defined by values that exceed a high threshold. Extreme value theory (EVT) gives a standard approximation for exceedances over such a threshold. Let $u(x)$ be a high threshold, which may depend on x, and define the exceedance

$$Z = Y - u(x), \tag{16}$$

considered conditionally on $Y > u(x)$. The conditional excess distribution is given by

$$F_u(z \mid x) = \Pr\big(Y - u(x) \leq z \mid Y > u(x), X = x\big), \qquad z \geq 0. \tag{17}$$

Under broad regularity conditions, $F_u(\cdot \mid x)$ is well-approximated by a generalized Pareto distribution (GPD) when $u(x)$ is large (Balkema and de Haan 1974; Pickands 1975; Coles 2001).

For $y > u(x)$, the corresponding GPD cumulative distribution function is given as

$$F_{\mathrm{GPD}}(y \mid u(x), \sigma(x), \xi) = \begin{cases} 1 - \left(1 + \xi \dfrac{y - u(x)}{\sigma(x)}\right)^{-1/\xi}, & \xi \neq 0, \\ 1 - \exp\left(-\dfrac{y - u(x)}{\sigma(x)}\right), & \xi = 0, \end{cases} \tag{18}$$

with support $y \geq u(x)$ when $\xi \geq 0$, and $u(x) \leq y \leq u(x) - \sigma(x)/\xi$ when $\xi < 0$. The corresponding density function is given by

$$f_{\mathrm{GPD}}(y \mid u(x), \sigma(x), \xi) = \begin{cases} \dfrac{1}{\sigma(x)} \left(1 + \xi \dfrac{y - u(x)}{\sigma(x)}\right)^{-1/\xi - 1}, & \xi \neq 0, \\ \dfrac{1}{\sigma(x)} \exp\left(-\dfrac{y - u(x)}{\sigma(x)}\right), & \xi = 0. \end{cases} \tag{19}$$

The associated quantile function is given by

$$Q_{\mathrm{GPD}}(\tau \mid u(x), \sigma(x), \xi) = \begin{cases} u(x) - \sigma(x) \log(1 - \tau), & \xi = 0, \\ u(x) + \dfrac{\sigma(x)}{\xi}\Big((1 - \tau)^{-\xi} - 1\Big), & \xi \neq 0, \end{cases} \qquad 0 < \tau < 1. \tag{20}$$

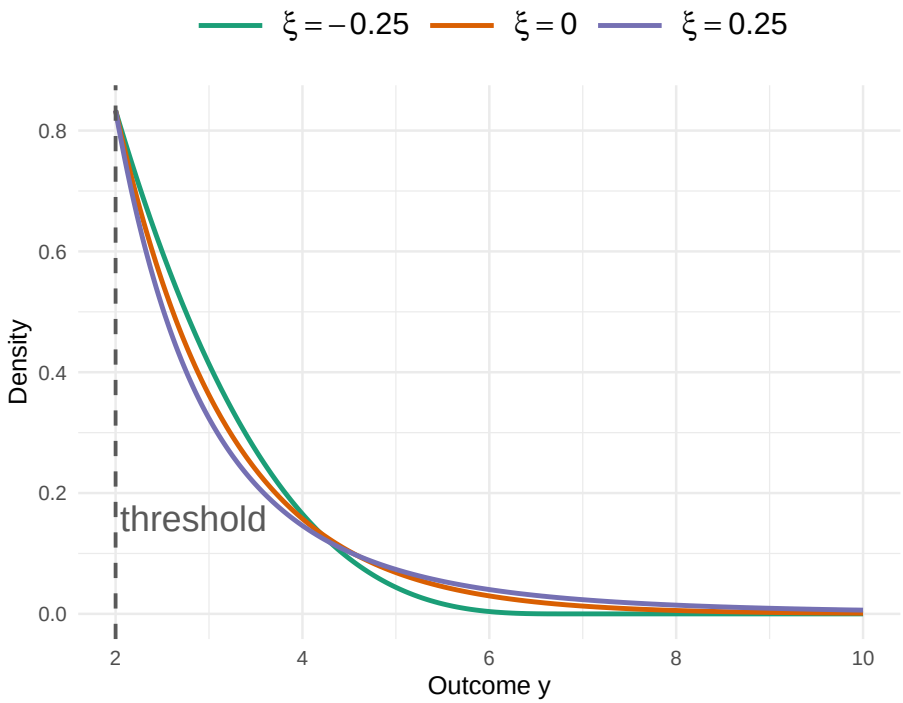

Figure 2: GPD tail density above threshold for multiple shape parameters.

Figure 2, illustrates the GPD density for different shape parameters ξ with a fixed scale parameter σ and threshold u. The shape parameter ξ controls the tail heaviness: $\xi < 0$ corresponds to a light tail with a finite upper bound, $\xi = 0$ corresponds to an exponential tail, and $\xi > 0$ corresponds to a heavy tail with infinite support. The GPD provides a flexible family for modeling exceedances above a high threshold, making it the tail component used in the spliced model.

7.2. Dirichlet Process Mixture Model for Conditional Densities

In this package, conditional density of Y given X below a specific threshold ($u(x)$) is modeled using a DPM regression model. Let $k(\cdot \mid x; \theta)$ be a user-chosen kernel family indexed by parameter θ. The hierarchical specification is

$$Y \mid X = x, \theta \sim k(\cdot \mid x; \theta), \qquad \theta \mid H \sim H, \qquad H \mid \kappa, H_0 \sim \mathrm{DP}(\kappa, H_0), \tag{21}$$

where H is an unknown mixing distribution on the kernel-parameter space, H_0 is a base measure, and $\kappa > 0$ is a concentration parameter (Ferguson 1973; Lo 1984). Marginalizing H yields an (almost surely) discrete mixing distribution and an infinite mixture representation:

$$f_{\mathrm{DP}}(y \mid x) = \int k(y \mid x; \theta)\, dH(\theta) = \sum_{j=1}^{\infty} w_j\, k(y \mid x; \theta_j), \qquad w_j \geq 0, \qquad \sum_{j=1}^{\infty} w_j = 1. \tag{22}$$

One representation for the weights is given by Sethuraman's stick-breaking construction

(Sethuraman 1994):

$$w_1 = V_1, \qquad w_j = V_j \prod_{\ell<j}(1 - V_\ell), \qquad V_j \overset{\text{iid}}{\sim} \text{Beta}(1, \kappa), \qquad \theta_j \overset{\text{iid}}{\sim} H_0. \tag{23}$$

An equivalent characterization is obtained by marginalizing over H, yielding the Blackwell–MacQueen Pólya urn representation. In particular, the conditional distribution of a parameter value given the others can be written as

$$\theta_i \mid \{\theta_j\}_{j\neq i} \sim \frac{\kappa}{\kappa + n} H_0(\cdot) + \frac{1}{\kappa + n} \sum_{j=1}^{n} \delta_{\theta_j}(\cdot), \tag{24}$$

where $\delta_{\theta_j}(\cdot)$ is a point mass assigned to θ_j . This induces a clustering structure often described via the Chinese restaurant process (CRP) (Blackwell and MacQueen 1973; Neal 2000). Both stick-breaking and CRP representations are widely used for posterior simulation in DPM models (Escobar and West 1995; Neal 2000).

7.3. Supplementary clustering details

This subsection records the fuller clustering constructions summarized in Section 2.2. When the predictor-indexed mixing measure in Eq. (11) is discrete, latent allocations induce a random partition of the observations. Introduce latent cluster-label variables z_i and component parameters $\{\theta_j\}_{j\geq 1}$ such that

$$y_i \mid z_i, \{\theta_j\}, x_i \sim k(\cdot \mid x_i, \theta_{z_i}) \,. \tag{25}$$

The induced partition $\Pi = \{z_1, \ldots, z_n\}$ is the object summarized by the clustering tools.

Predictor dependence structures. Starting from Eq. (11), the package supports three DDP-style dependence patterns:

- **Weight dependence only:** predictors modulate mixture weights while atoms are shared across x,

$$f(y \mid x) \;=\; \sum_{j=1}^{\infty} w_j(x)\, k(y \mid \theta_j) \,. \tag{26}$$

  This is appropriate when covariates change cluster prevalence but not the within-cluster kernel shape, and includes logistic stick-breaking constructions (Ren *et al.* 2011).

- **Atom dependence only:** predictors enter through component-specific kernel parameters while weights remain global,

$$f(y \mid x) \;=\; \sum_{j=1}^{\infty} w_j\, k(y \mid x, \theta_j) \,. \tag{27}$$

  Here clusters correspond to distinct regression surfaces or parameter regimes while w_j retains the interpretation of global prevalence (MacEachern 1999, 2000; De Iorio *et al.* 2004).

- **Weight and atom dependence:** predictors modulate both prevalence and within-cluster behavior,

$$f(y \mid x) \;=\; \sum_{j=1}^{\infty} w_j(x)\, k(y \mid x, \theta_j)\,. \tag{28}$$

  This is the most flexible DDP-style specification and allows both selection probabilities and cluster-specific conditional structure to vary with covariates (Quintana *et al.* 2022).

Label-invariant posterior summaries. Posterior clustering is reported through partition functionals rather than raw MCMC labels. The posterior similarity matrix (PSM) has entries

$$S_{i\ell} \;=\; \Pr(z_i = z_\ell \mid \text{data}), \qquad i, \ell = 1, \ldots, n, \tag{29}$$

estimated by Monte Carlo averages over retained draws. A representative partition is obtained by selecting the sampled partition whose adjacency matrix is closest to the PSM under squared loss (Dahl 2006); this yields a stable point estimate while preserving the PSM as the primary uncertainty summary.

7.4. Supplementary causal notation and identification details

This subsection expands the abbreviated causal background in Section 2.3. The reported causal contrasts themselves are listed in Table 1; here we define the underlying conditional and standardized functionals.

Arm-specific distributions and standardized functionals. Building from the observed-outcome identity in Eq. (12), the treatment-specific conditional distribution functions are

$$F_a(y \mid x) = \Pr\,\{Y(a) \le y \mid X = x\}, \qquad a \in \{0, 1\}. \tag{30}$$

with corresponding conditional density $f_a(y \mid x)$. Conditional quantiles are defined by

$$Q_a(\tau \mid x) = \inf\{y :\, F_a(y \mid x) \ge \tau\}, \qquad \tau \in (0, 1). \tag{31}$$

Conditional means are

$$\mu_a(x) = \mathrm{E}\{Y(a) \mid X = x\} = \int y\, f_a(y \mid x)\, dy. \tag{32}$$

Population-standardized and treated-standardized distribution functions are

$$F_a^{\mathrm{m}}(y) = \mathrm{E}\{F_a(y \mid X)\}, \qquad F_a^{\mathrm{t}}(y) = \mathrm{E}\{F_a(y \mid X) \mid A = 1\}. \tag{33}$$

The corresponding quantiles are

$$Q_a^{\mathrm{m}}(\tau) = \inf\{y :\, F_a^{\mathrm{m}}(y) \ge \tau\}, \qquad Q_a^{\mathrm{t}}(\tau) = \inf\{y :\, F_a^{\mathrm{t}}(y) \ge \tau\}. \tag{34}$$

and marginal means are

$$\mu_a^{\mathrm{m}} = \mathrm{E}\{\mu_a(X)\}, \qquad \mu_a^{\mathrm{t}} = \mathrm{E}\{\mu_a(X) \mid A = 1\}. \tag{35}$$

Identification and propensity-score augmentation. Identification depends on the study design. In randomized studies, randomization together with positivity identifies the treatment-specific distributions. In observational studies, we additionally rely on SUTVA, consistency, conditional ignorability,

$$(Y(0), Y(1)) \perp A \mid X, \tag{36}$$

and overlap

$$0 < \Pr(A = 1 \mid X = x) < 1 \tag{37}$$

for the covariate profiles of interest. Under these conditions, the treatment-specific conditional distributions are identified from the observed data:

$$F_a(y \mid x) = \Pr(Y \leq y \mid A = a, X = x), \qquad a \in \{0, 1\}. \tag{38}$$

The propensity score is

$$\rho(x) = \Pr(A = 1 \mid X = x) \tag{39}$$

In the observational workflow, the package first estimates $\rho(x)$ using only (A, X) and then augments the outcome model covariates with the estimated score:

$$\widehat{\rho}(x) \approx \rho(x), \qquad r(x) = \left(x^{\mathsf{T}}, \widehat{\rho}(x)\right)^{\mathsf{T}}. \tag{40}$$

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